EXPLORING THE N-VECTOR MODEL

DIMENSIONALITY OF SPINS

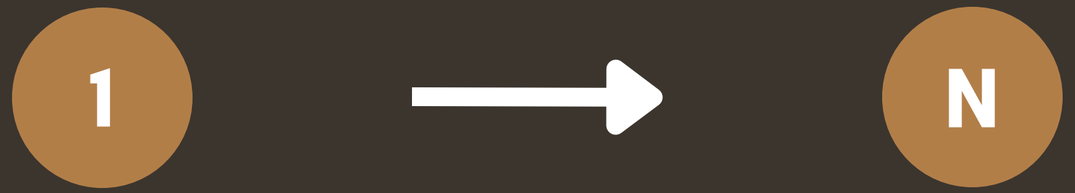
PRESENTING:
JAKUB KWASNIAK
MEHMET EMRE DURMUS
JAKUB DOMANSKI

N-VECTOR MODEL

Also known as the O(n) model, developed by Harry Eugene Stanley in 1968 as a generalisation of the Lenz-Ising model to n-dimensional spins [1].

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Spin lattice models provide an effective framework for the study of ferromagnetism.

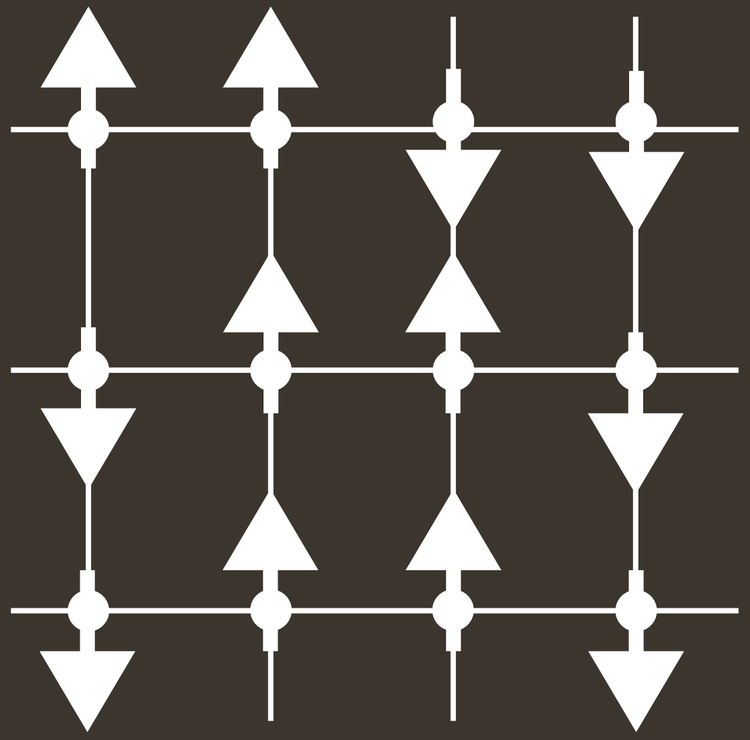
Important within the Standard Model and integrable systems.

RESEARCH QUESTION



How does the dimensionality of the spins affect the relevant observables, i.e. energy, magnetisation, specific heat?

LENZ-ISING MODEL



Describes the dynamics of interacting classical spins on a lattice [4].

$$H(\sigma) = -\sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j - \mu \sum_i h_i\sigma_i$$

W. Lenz (1920) [2], E. Ising (1924) [3]

KEY ASSUMPTIONS

1. No external magnetic field influence.

2. Constant interaction strength between neighbouring spins.

$$\Longrightarrow H(\sigma) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j$$

METROPOLIS-HASTINGS ALGORITHM

Markov Chain Monte Carlo (MCMC) method, using the Monte Carlo estimator to compute observables [5].

$$\tilde{F} = \frac{1}{N} \sum_{i \in \{X_n\}} \frac{f(i)}{\pi_i}$$

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By choosing an appropriate function and the distribution, we may evaluate quantities of interest, e.g. the total energy.

$$\Longrightarrow \quad \tilde{E} = \frac{1}{N} \sum_{i \in \{X_n\}} H(i)$$

METROPOLIS-HASTINGS ALGORITHM

Our Markov Chain for the Ising Model consists of spin degrees of freedom

N. Metropolis et al. (1953) [6], W. Hastings (1970) [7]

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$$\sigma_j$$

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$$\sigma_j$$

$$\downarrow$$

Proposal: Flip the spin

$$\sigma_i$$

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Our Markov Chain for the Ising Model consists of spin degrees of freedom

$$\sigma_j$$

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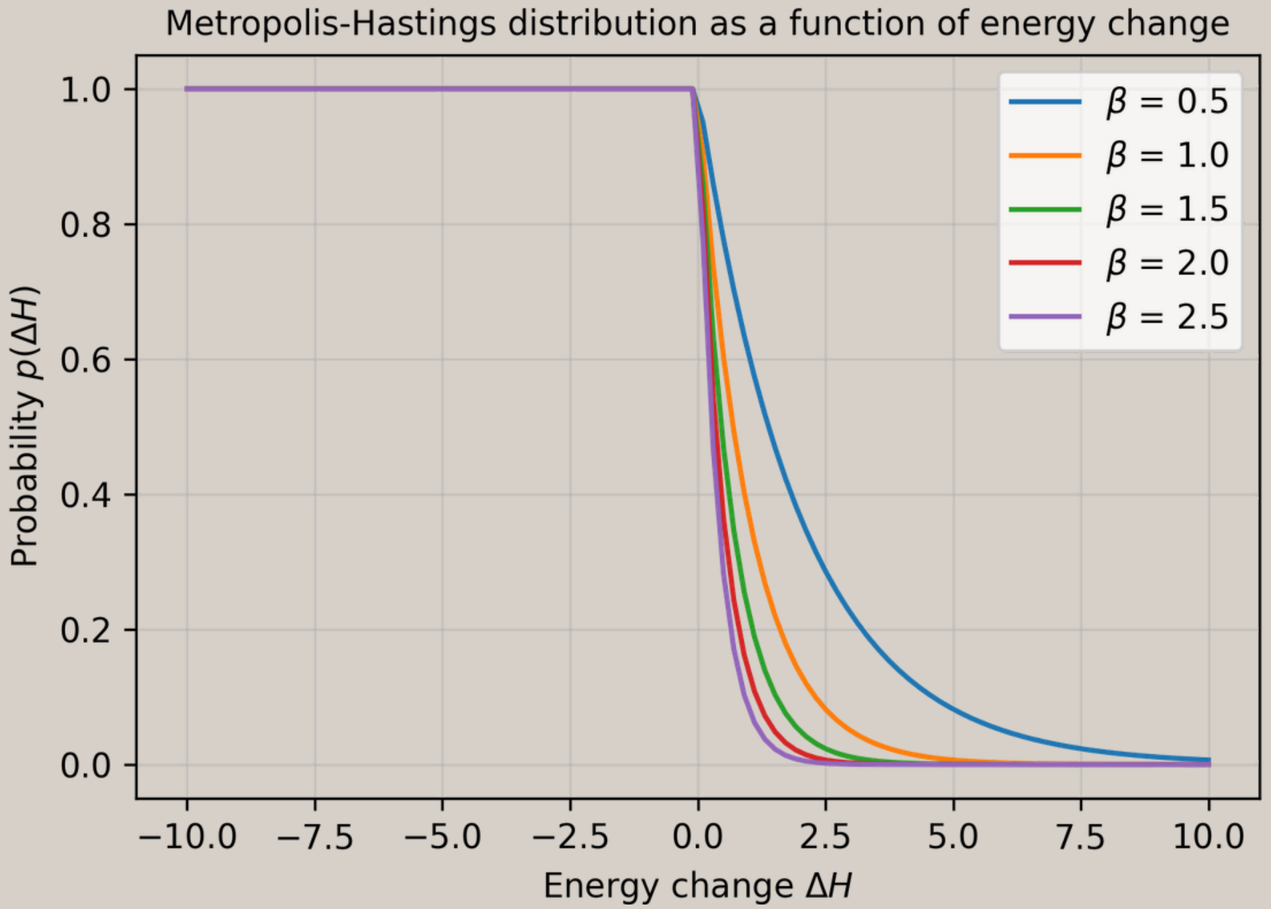
Reject $1 - A_{ij}$ $\longleftarrow$ σ_i $\longrightarrow$ Accept A_{ij}

N. Metropolis et al. (1953) [6], W. Hastings (1970) [7]

METROPOLIS-HASTINGS ALGORITHM

- Metropolis-Hastings distribution

$$A = \begin{cases} e^{-\beta \Delta H} & \text{if } \Delta H > 0 \\ 1 & \text{else} \end{cases}$$



IMPLEMENTATION

Reproduced the Metropolis-Hastings algorithm in Julia programming language.

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1. Generate a MxN lattice of d-dimensional spins.

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3. Calculate the energy difference between the configurations.

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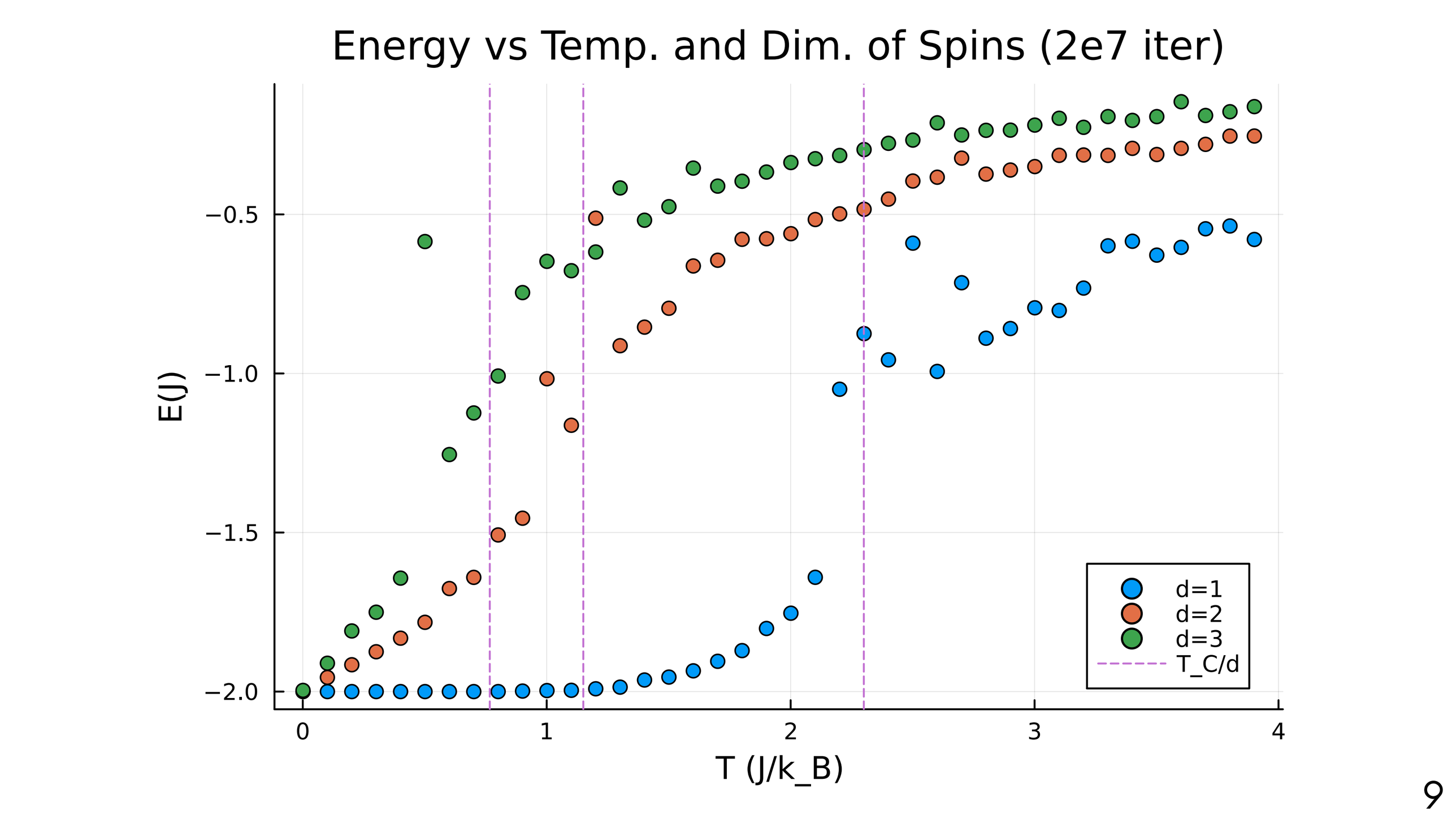
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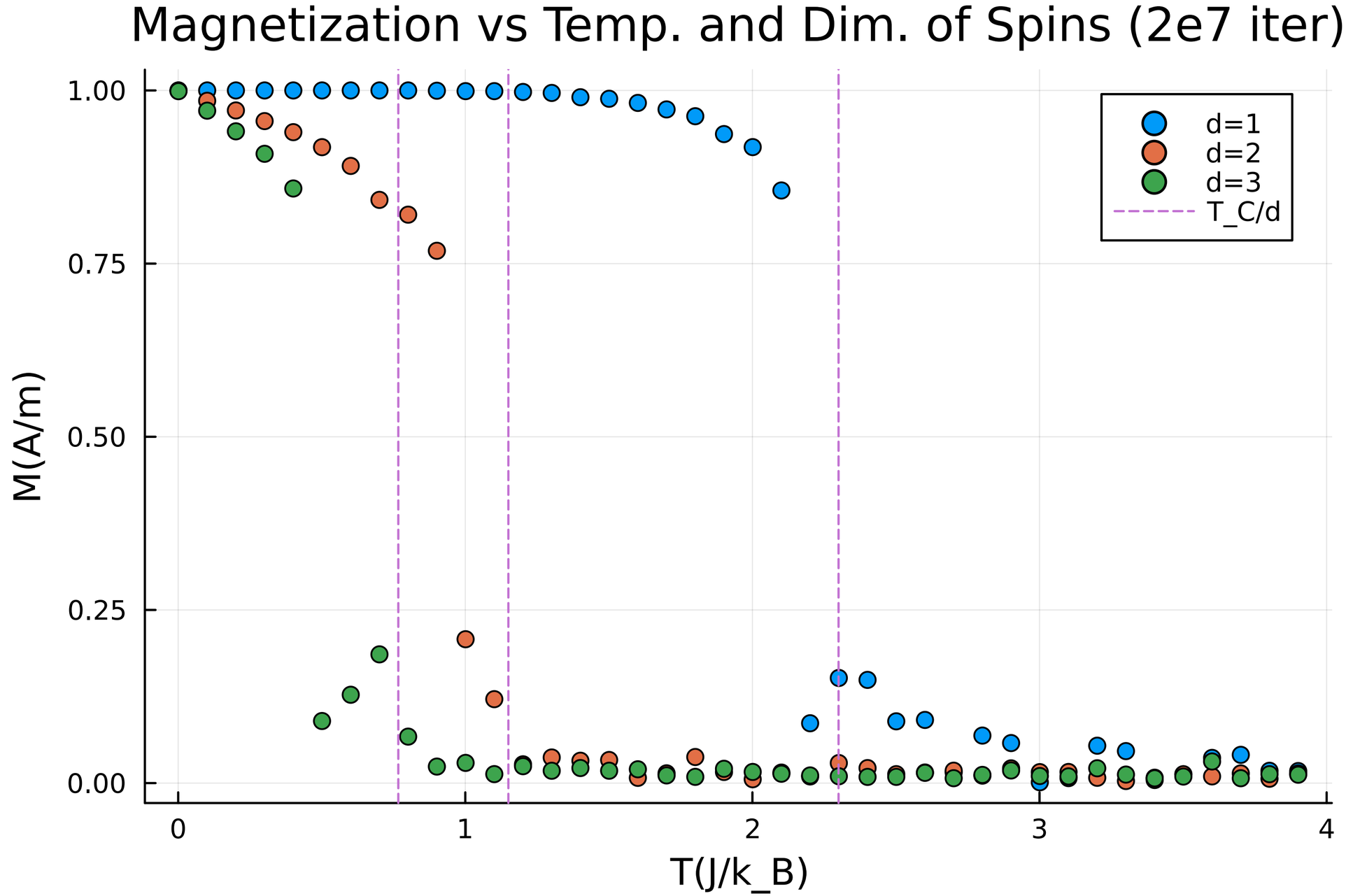
1. Generate a MxN lattice of d-dimensional spins.
2. Randomly select a lattice site, choose a random spin.
3. Calculate the energy difference between the configurations.
4. Compare to acceptance matrix A.

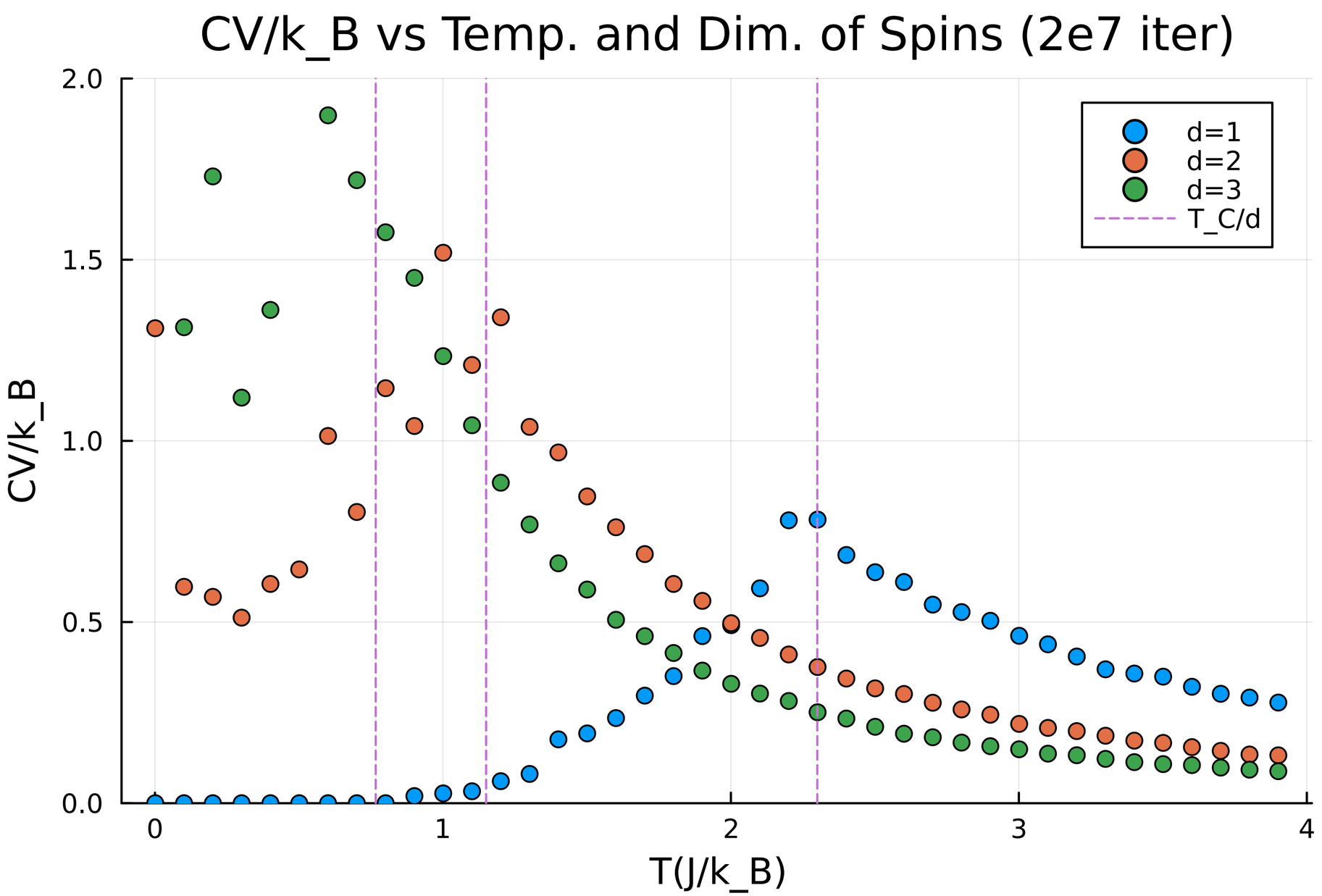
IMPLEMENTATION

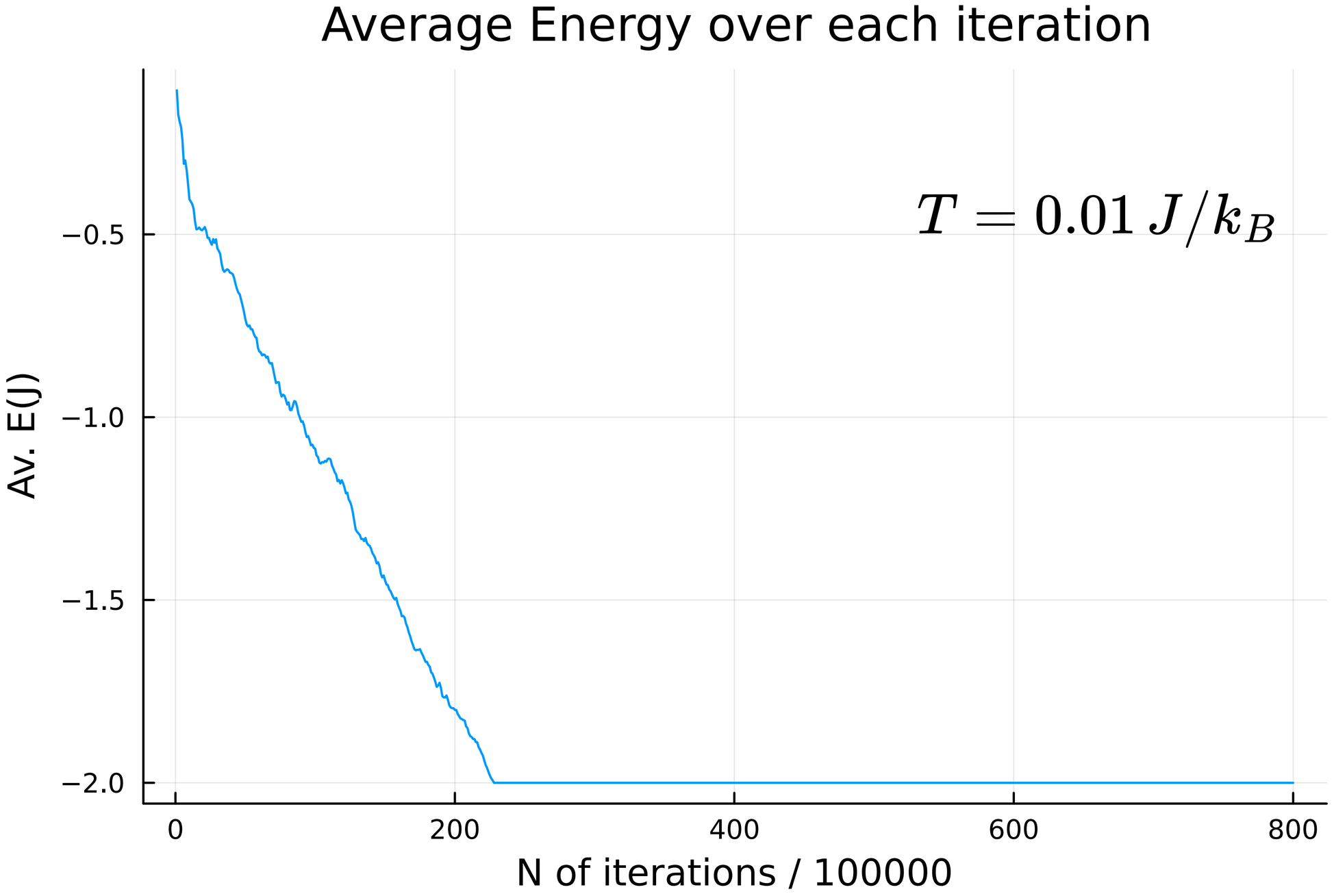
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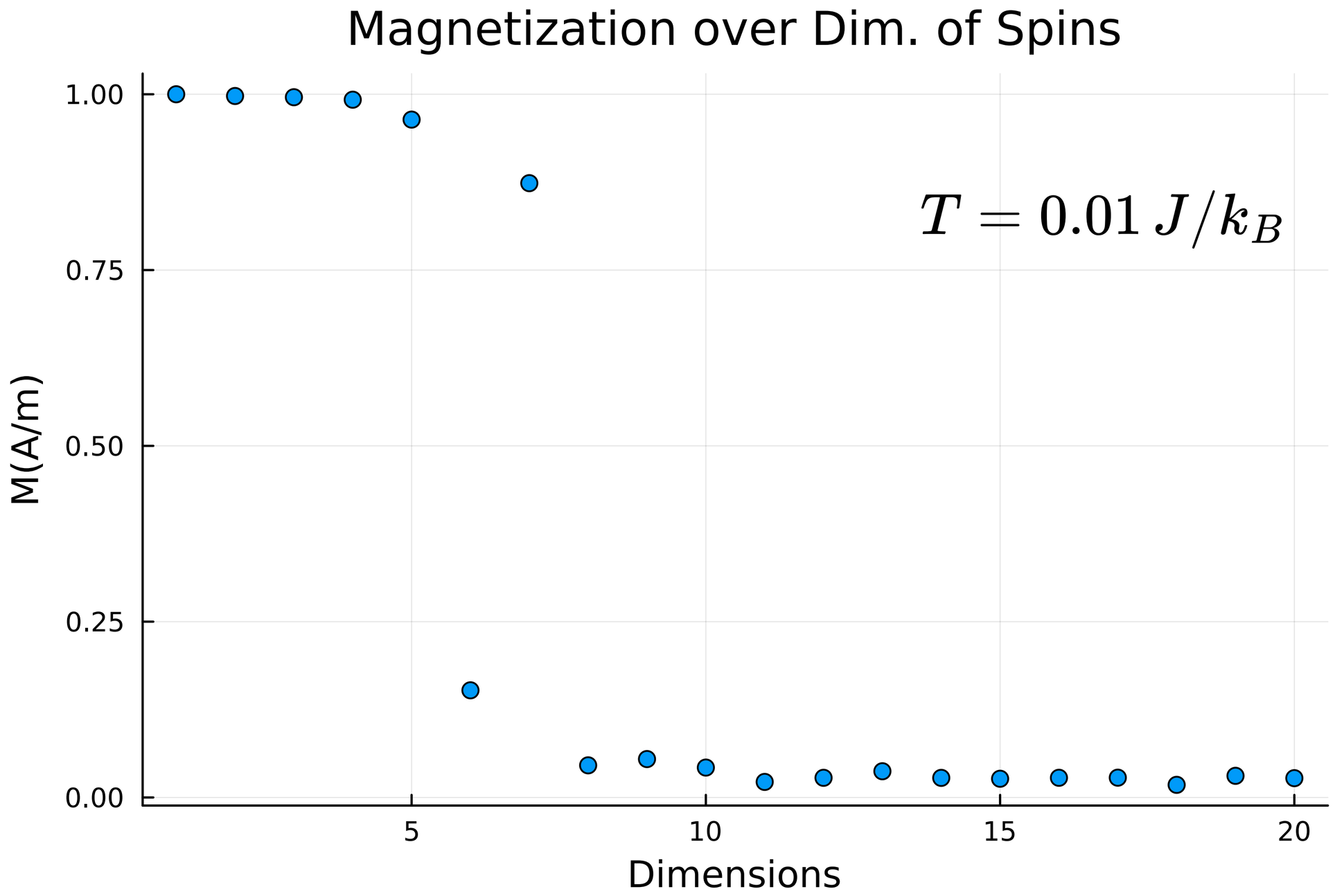
1. Generate a MxN lattice of d-dimensional spins.
2. Randomly select a lattice site, choose a random spin.
3. Calculate the energy difference between the configurations.
4. Compare to acceptance matrix A.
5. Repeat.


Energy vs Temp. and Dim. of Spins (2e7 iter)
E(J)
−0.5
−1.0
−1.5
−2.0
0
1
2
3
4
T (J/k_B)
d=1
d=2
d=3
T_C/d

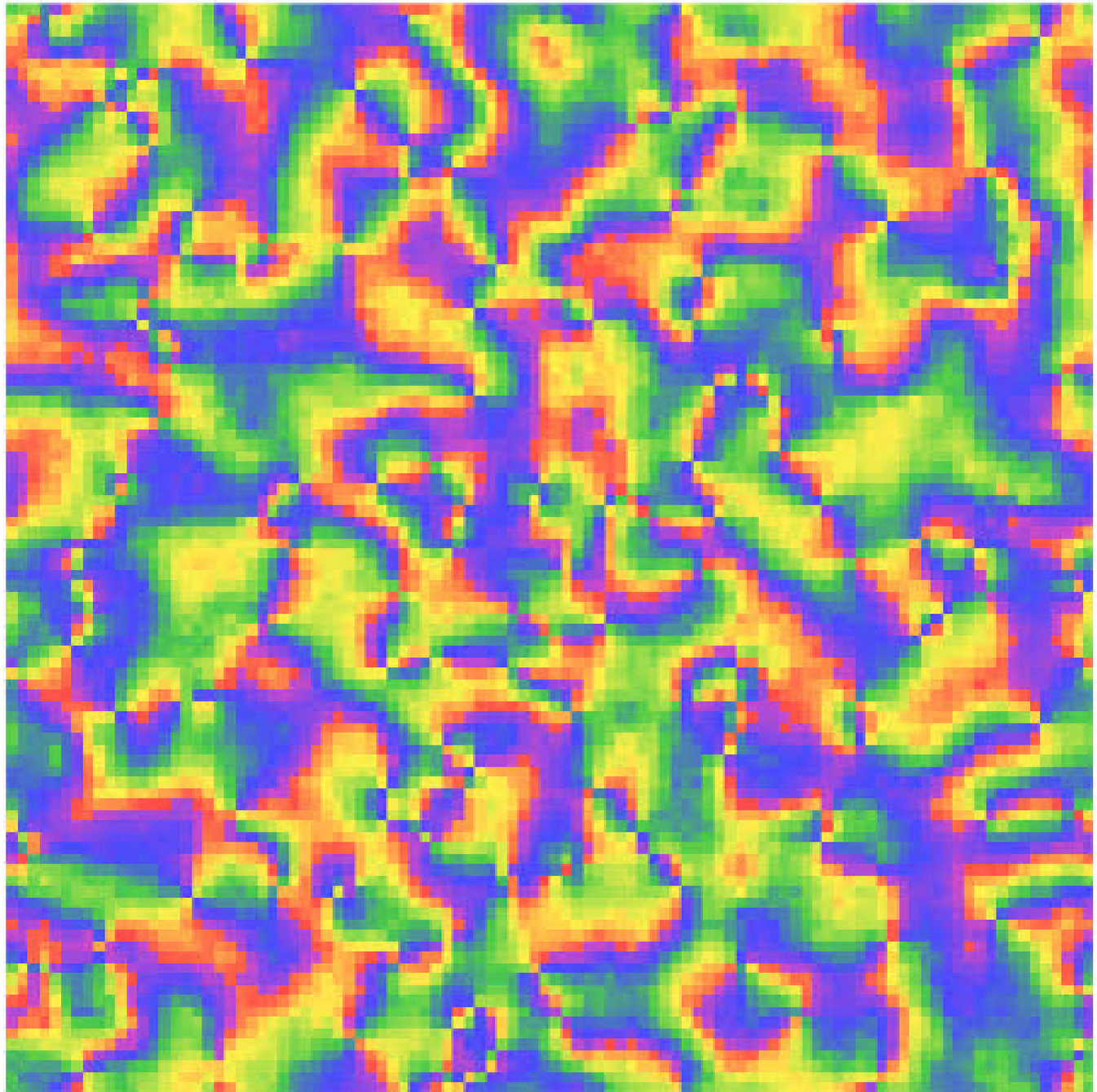

Magnetization vs Temp. and Dim. of Spins (2e7 iter)
M(A/m)
T(J/k_B)
1.00
0.75
0.50
0.25
0.00
0
1
2
3
4
d=1
d=2
d=3
T_C/d


CV/k_B vs Temp. and Dim. of Spins (2e7 iter)
CV/k_B
T(J/k_B)
2.0
1.5
1.0
0.5
0.0
0
1
2
3
4
d=1
d=2
d=3
T_C/d


Average Energy over each iteration
$T = 0.01\, J/k_B$
Av. E(J)
−0.5
−1.0
−1.5
−2.0
0
200
400
600
800
N of iterations / 100000


Magnetization over Dim. of Spins
$T = 0.01\, J/k_B$
M(A/m)
1.00
0.75
0.50
0.25
0.00
5
10
15
20
Dimensions

θ evolution for T / kB = 0.01



CONCLUSIONS

1. Increasing dimensions implies a decrease in the magnetisation.

2. Small temperature regime requires longer computation time.

3. Disordered behaviour more prominent in higher dimensions.

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[1] H. E. Stanley, “Dependence of Critical Properties on Dimensionality of Spins,” Phys. Rev. Lett., vol. 20, no. 589, Mar. 1968. doi: 10.1103/PhysRevLett.20.589.

[2] W. Lenz, “Beitrag zum Verständnis der magnetischen Erscheinungen in festen Körpern,” Z. Phys., vol. 21, pp. 613–615, 1920.

[3] E. Ising, “Beitrag zur Theorie des Ferromagnetismus,” Z. Phys., vol. 31, pp. 253–258, Feb. 1925.

[4] B. A. Cipra, “An Introduction to the Ising Model,” Am. Math. Mon., vol. 94, no. 10, pp. 937–959, Dec. 1987. doi: 10.2307/2322600.

[5] M. Kastner, “Monte Carlo methods in statistical physics: Mathematical foundations and strategies,” Commun. Nonlinear Sci. Numer. Simul., vol. 15, no. 6, pp. 1589–1602, Jun. 2010. doi: 10.1016/j.cnsns.2009.06.011.

[6] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, “Equation of State Calculations by Fast Computing Machines,” J. Chem. Phys., vol. 21, no. 6, pp. 1087–1092, Jun. 1953.

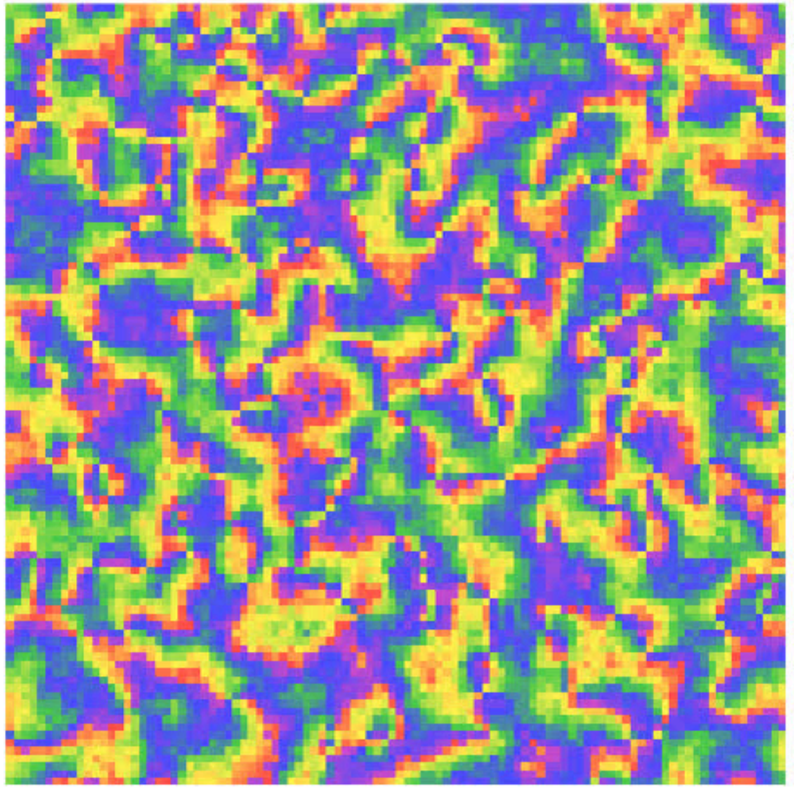
[7] W. K. Hastings, “Monte Carlo Sampling Using Markov Chains and Their Applications,” Biometrika, vol. 57, no. 1, pp. 97–109, Apr. 1970. doi: 10.2307/2334940.

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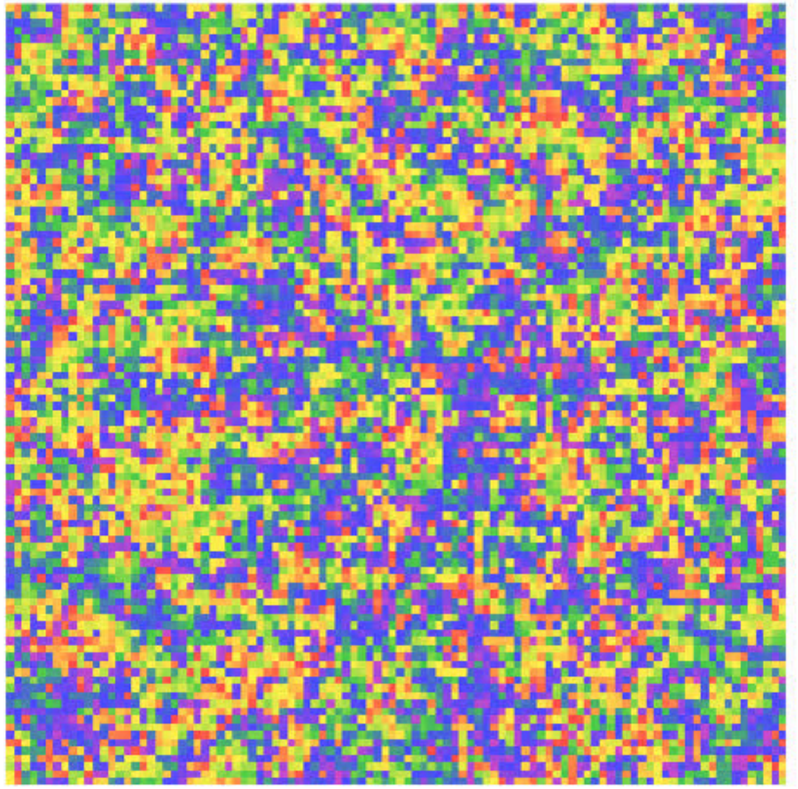
For more details about the project, including the source code and a review document on the theoretical background, feel free to visit:

https://github.com/jakub0628/PPM-Numerical-Methods-Lattice-Model

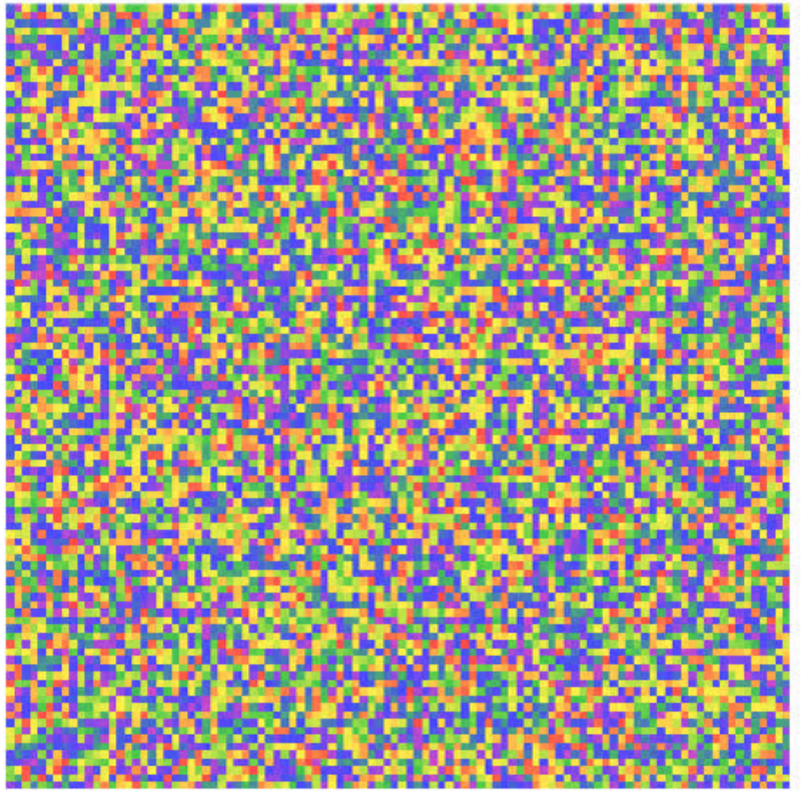
θ evolution for T / kB = 0.1



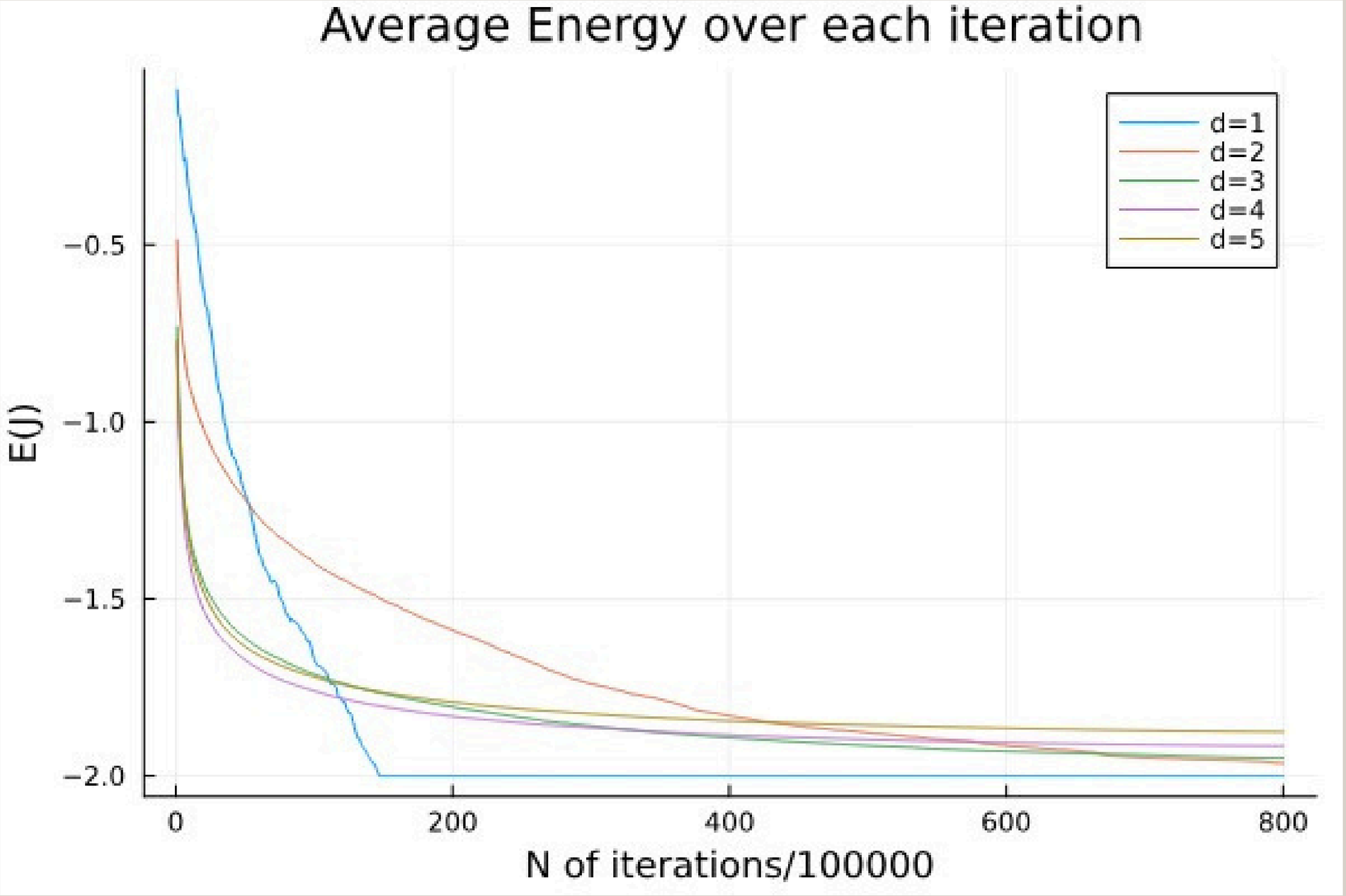
θ evolution for T / kB = 1.0

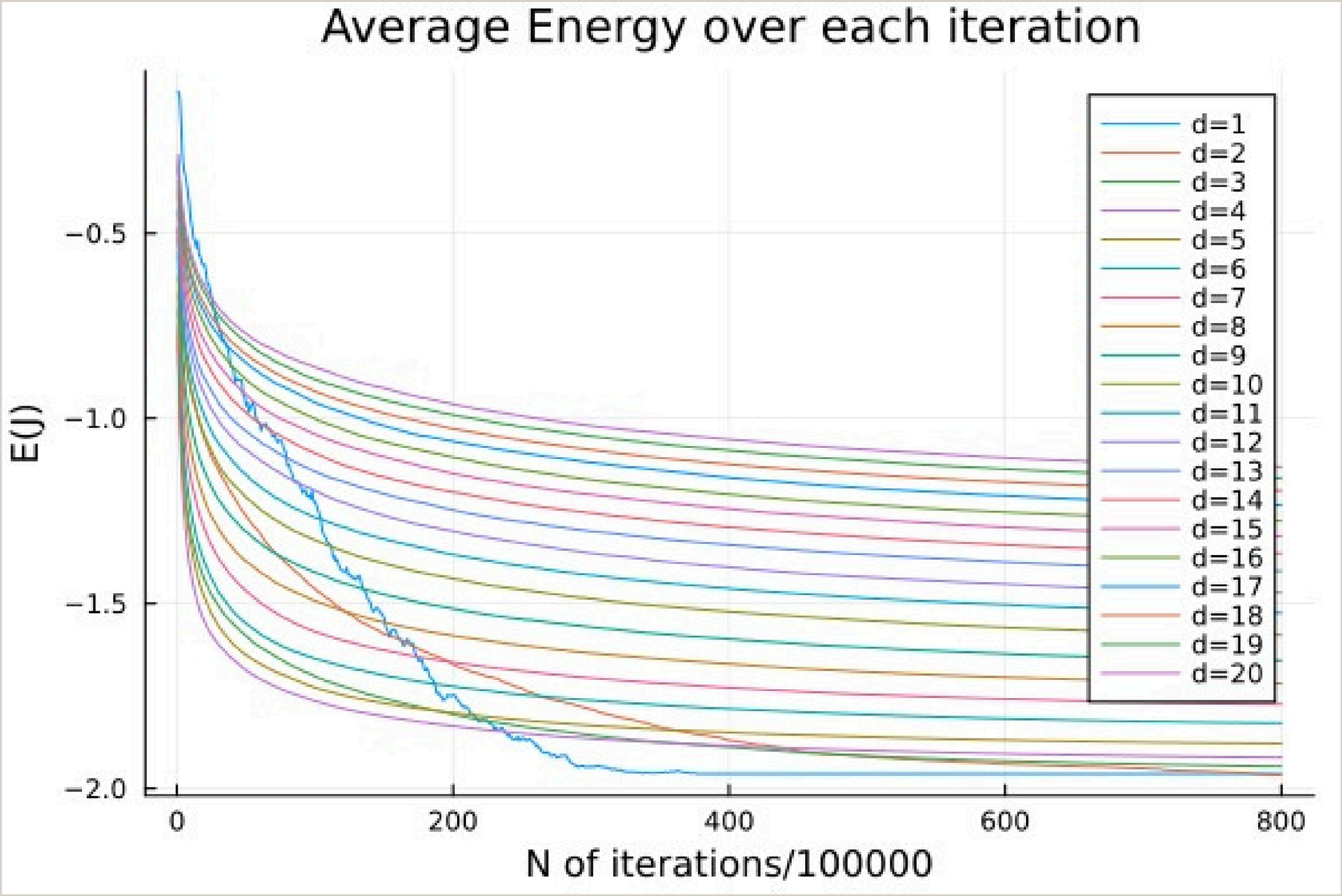


θ evolution for T / kB = 10.0



BACKUP SLIDES


Average Energy over each iteration
d=1
d=2
d=3
d=4
d=5
−0.5
−1.0
−1.5
−2.0
E(J)
0
200
400
600
800
N of iterations/100000


Average Energy over each iteration
E(J)
−0.5
−1.0
−1.5
−2.0
0
200
400
600
800
N of iterations/100000
d=1
d=2
d=3
d=4
d=5
d=6
d=7
d=8
d=9
d=10
d=11
d=12
d=13
d=14
d=15
d=16
d=17
d=18
d=19
d=20

METROPOLIS-HASTINGS ALGORITHM

Markov Chain Monte Carlo (MCMC) method, using the Monte Carlo estimator to compute observables [5].

Markov Chain: $\{X_n\} = \{\lambda, T\lambda, T^2\lambda, \dots T^{N-1}\lambda\}$

Sequence of random variables computed by applying a given transition matrix T to the initial distribution λ.

X_{i+1} depends only on $X_i \Longrightarrow$ Markov Chain "has no memory"

Monte Carlo Estimator function

$$\tilde{F} = \frac{1}{N} \sum_{i \in \{X_n\}} \frac{f(i)}{\pi_i}$$

$\pi = (\pi_i : i \in S)$ - "stationary" distribution

N - length of the Markov Chain

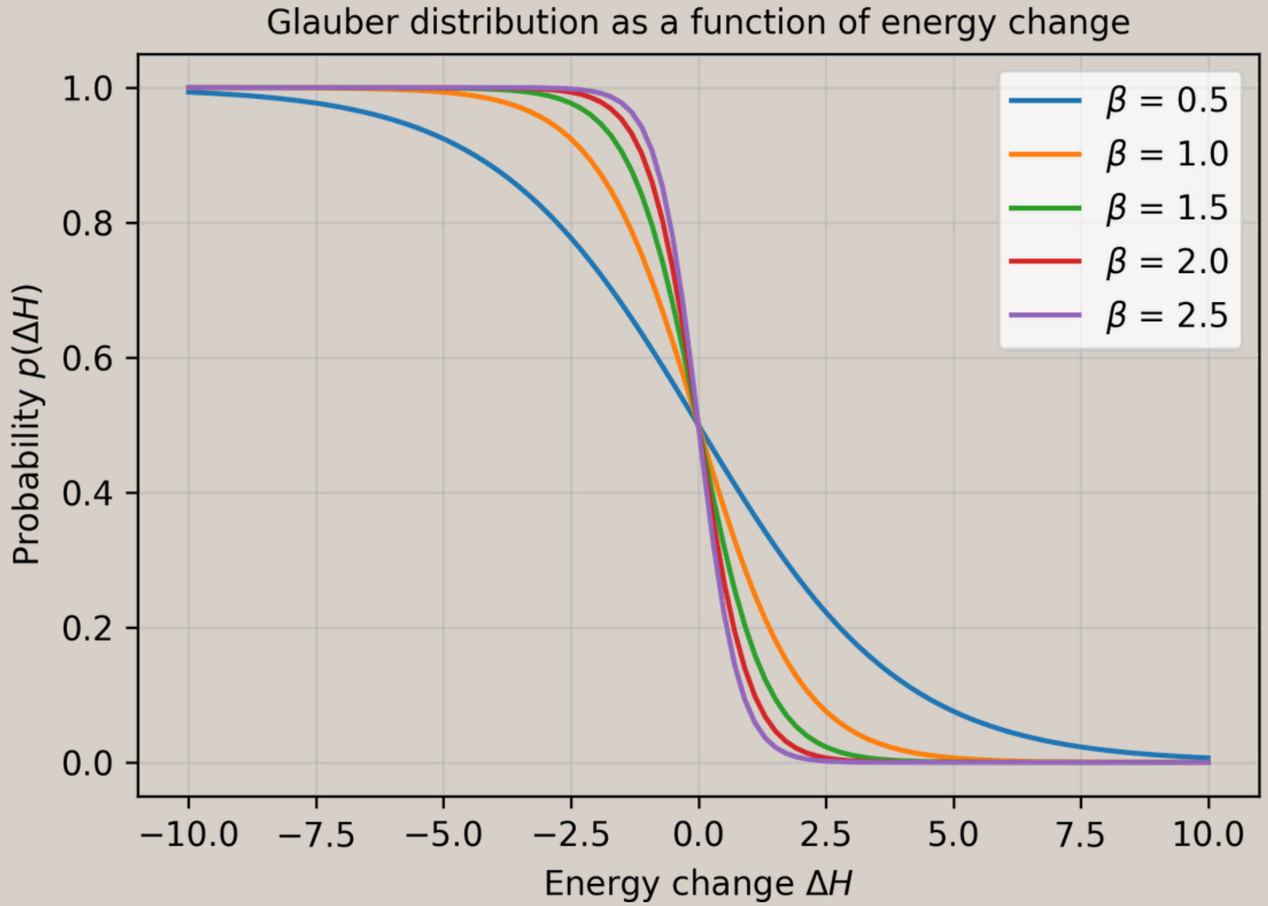
f - any function to be computed

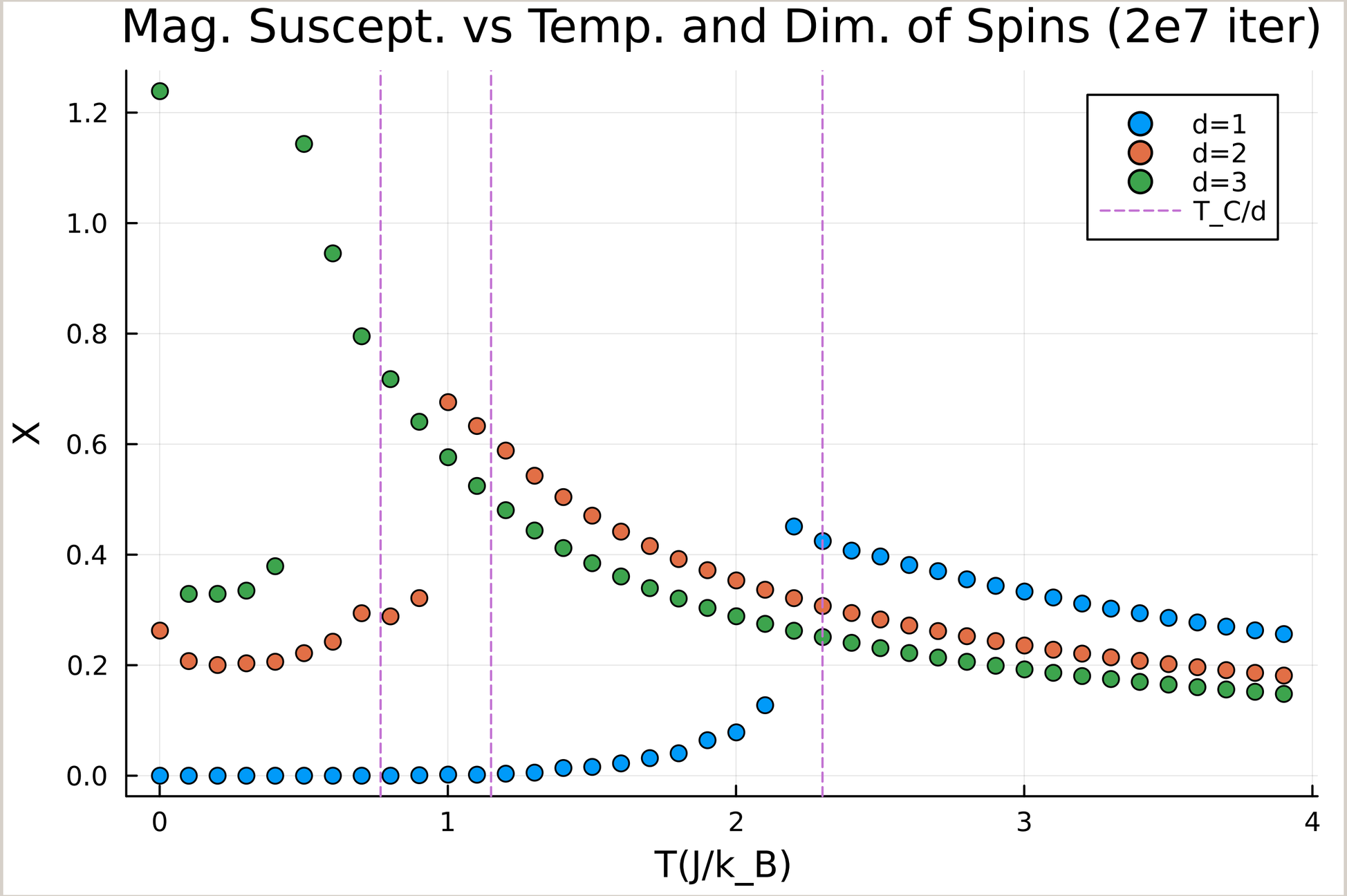
METROPOLIS-HASTINGS ALGORITHM

Two common choices of the acceptance matrix

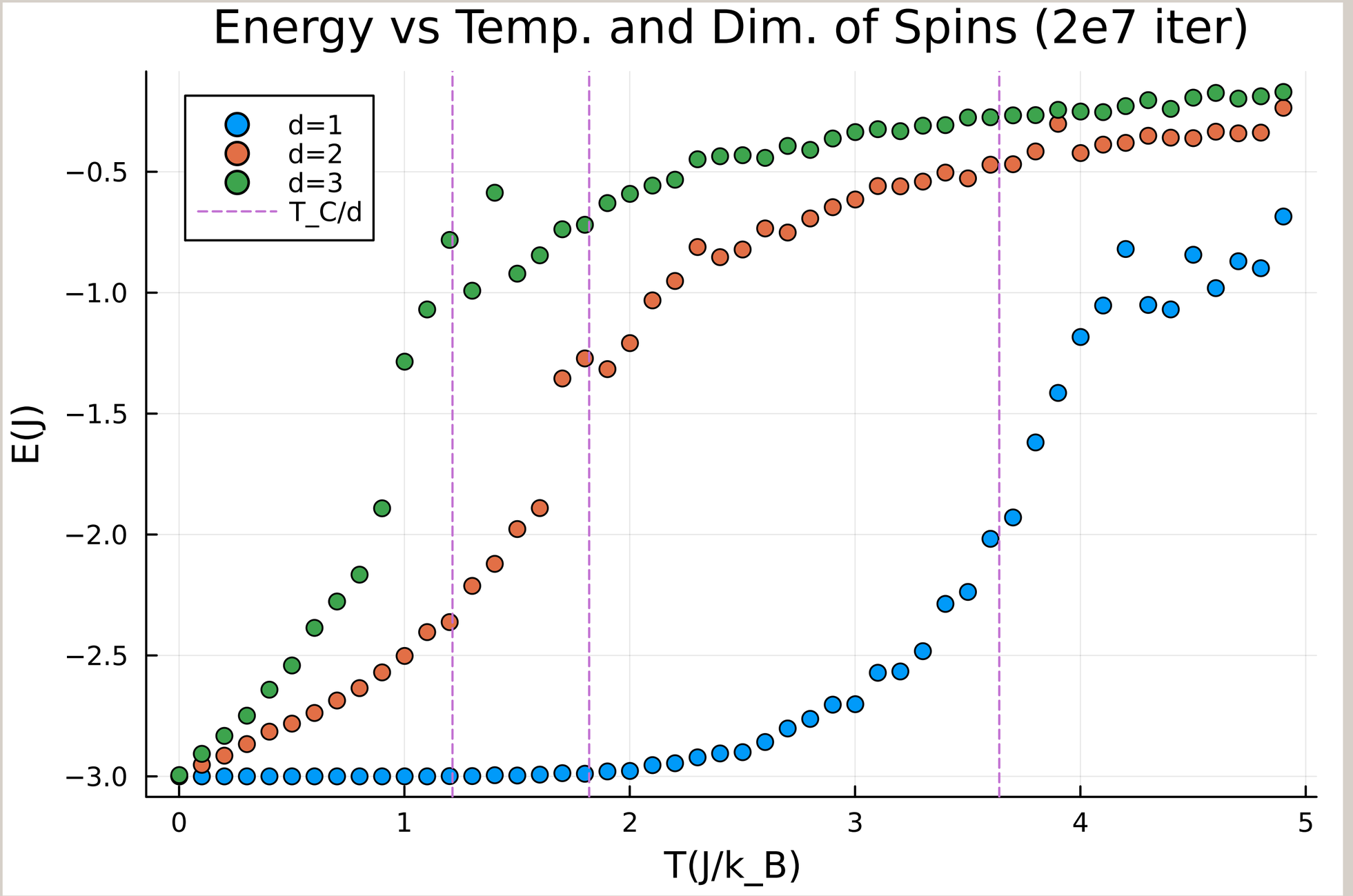
- Glauber distribution

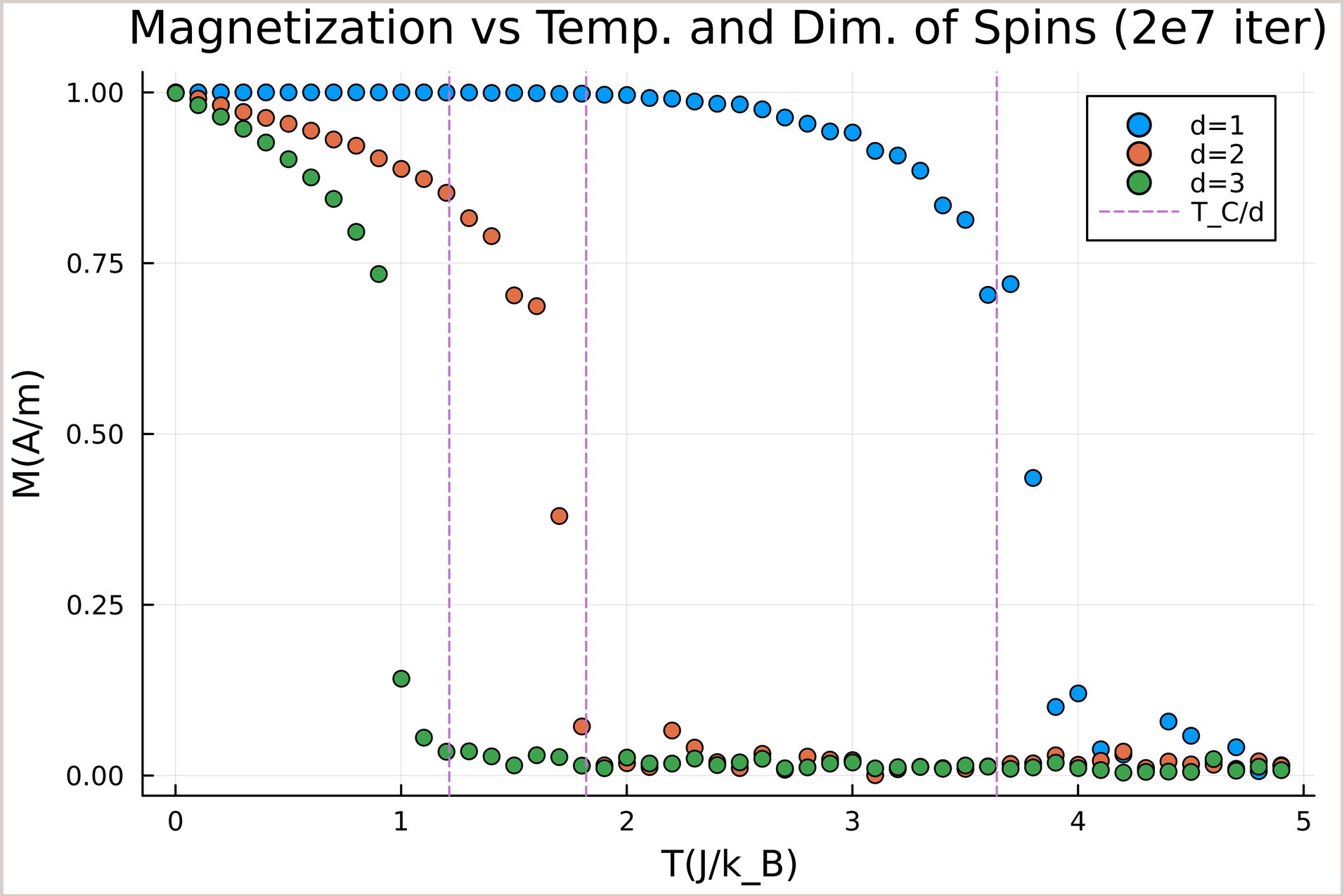
$$A = \begin{cases} (1 + e^{\beta \Delta H})^{-1} & \text{if } \Delta H \neq 0 \\ 1/2 & \text{else} \end{cases}$$

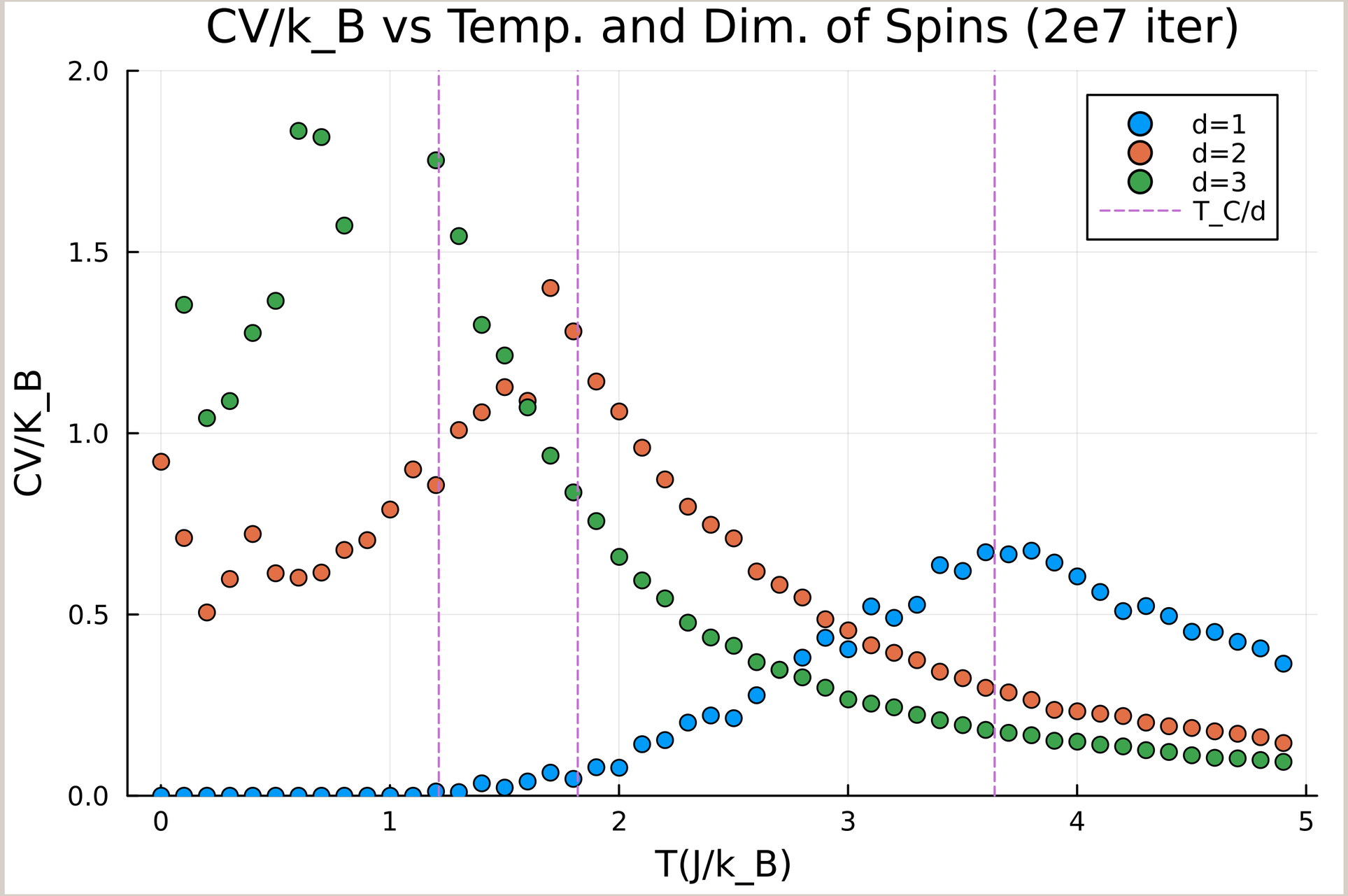



Mag. Suscept. vs Temp. and Dim. of Spins (2e7 iter)
d=1
d=2
d=3
T_C/d
χ
T(J/k_B)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4

TRIANGULAR LATTICE




Magnetization vs Temp. and Dim. of Spins (2e7 iter)
M(A/m)
T(J/k_B)
1.00
0.75
0.50
0.25
0.00
0
1
2
3
4
5
d=1
d=2
d=3
T_C/d


CV/k_B vs Temp. and Dim. of Spins (2e7 iter)
CV/K_B
T(J/k_B)
0.0
0.5
1.0
1.5
2.0
0
1
2
3
4
5
d=1
d=2
d=3
T_C/d